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ARTI 402 - Building a Basic Sentiment Analyzer with Python And Textblob

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Introduction

Understanding customer opinions has become an essential part of modern product development and service improvement. With the rapid growth of online shopping platforms such as Amazon, customers frequently express their experiences through written reviews, creating large amounts of valuable textual data. Analyzing this data manually is time-consuming and inefficient, which highlights the need for intelligent systems capable of automatically identifying sentiment within text.

Sentiment analysis provides a way to determine whether a piece of text expresses a positive, negative, or neutral emotion. By applying Natural Language Processing (NLP) techniques, machines can interpret human language and classify the feelings or attitudes conveyed in reviews. In this report, a simple sentiment analysis solution is developed using Python and the TextBlob library. TextBlob offers an accessible interface for calculating sentiment polarity, making it suitable for building lightweight and effective sentiment classifiers.

This technical report aims to explore how an AI-based sentiment analyzer can process Amazon product reviews and generate meaningful insights. The project focuses on reviews of unlocked mobile phones, a product category with a large amount of feedback. By analyzing these reviews, the system can support sellers and manufacturers by summarizing customer satisfaction, highlighting common concerns, and providing data-driven indications of overall product performance.

Problem statement

Online platforms such as Amazon host millions of product reviews generated by customers who share their experiences, opinions, and satisfaction levels. While this information is valuable for sellers and manufacturers, manually reading and analyzing large volumes of reviews is inefficient and impractical. Important insights—such as common complaints, positive feedback themes, and overall customer sentiment—can easily be overlooked.

The main problem is the lack of an automated system that can quickly classify the sentiment of customer reviews. Without such a tool, decision-makers struggle to understand customer satisfaction trends, identify issues early, or evaluate the general perception of their products. Therefore, an intelligent solution is needed to process textual data and categorize reviews as positive, negative, or neutral in a consistent and scalable manner.

AI Solution

1. Sentiment Analysis Function:

- A function called *sentimentAnalyzer(text)* was implemented to classify each review as Positive, Negative, or Neutral. This function uses TextBlob to compute the polarity score of the given tex.

2. Adding Sentiment and Polarity to the DataFrame:

- A new column named *Sentiment* was added to the DataFrame, storing the output of the *sentimentAnalyzer(text)* function for every review. This enabled the system to categorize all customer reviews based on their sentiment—Positive, Negative, or Neutral.

3. Using groupby:

- To obtain representative samples from the dataset, the *groupby* function was applied to group the reviews according to their assigned sentiment categories: Positive, Neutral, and Negative. From each sentiment group, the first review and its polarity score were extracted. This made it possible to provide clear examples showing how the AI model classified different types of feedback and helped evaluate the accuracy of the model's predictions.

4. Using Stopwords Filtering:

- To identify the main reasons behind the negative reviews, common English stopwords—such as "the," "with," and "this"—were removed. This ensured that only meaningful words contributing to the sentiment were analyzed.

Task 1:

```python
import pandas as pd
from textblob import TextBlob
```

```python
#task1
def sentimentAnalyzer(text):
    x = TextBlob(text)
    score = x.sentiment.polarity

    if score < -0.2:
        return "Negative"
    elif score > 0.2:
        return "Positive"
    else:
        return "Neutral"
```

figure (1): Task1

Task 2:

```python
#task2
print(sentimentAnalyzer('exciting '))
print(sentimentAnalyzer('happy'))
print(sentimentAnalyzer('good'))
print(sentimentAnalyzer('rich'))
print(sentimentAnalyzer('smile'))
print("-------------------")
print(sentimentAnalyzer('sad '))
print(sentimentAnalyzer('disappointed'))
print(sentimentAnalyzer('bad'))
print(sentimentAnalyzer('poor'))
print(sentimentAnalyzer('anger'))
print("-------------------")
print(sentimentAnalyzer('food '))
print(sentimentAnalyzer('animal '))
```

```
Positive
Positive
Positive
Positive
Positive
-------------------
Negative
Negative
Negative
Negative
Negative
-------------------
Neutral
Neutral
```

figure (2): Task2

Task 3:

125.8 ميغابايت 4 ديسمبر أنا Amazon_Unlocked_Mobile.csv

```
#task3
path = "//Amazon_Unlocked_Mobile.csv"
data = pd.read_csv(path)
data
```

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes
0	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	5	I feel so LUCKY to have found this used (phone...	1.0
1	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	4	nice phone, nice up grade from my pantach revu...	0.0
2	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	5	Very pleased	0.0
3	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	4	It works good but it goes slow sometimes but i...	0.0
4	"CLEAR CLEAN ESN" Sprint EPIC 4G Galaxy SPH-D7...	Samsung	199.99	4	Great phone to replace my lost phone. The only...	0.0
...	...	...	...	...	...	...
413835	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	5	another great deal great price	0.0
413836	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	3	Ok	0.0
413837	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	5	Passes every drop test onto porcelain tile!	0.0
413838	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	3	I returned it because it did not meet my needs...	0.0
413839	Samsung Convoy U640 Phone for Verizon Wireless...	Samsung	79.95	4	Only downside is that apparently Verizon no lo...	0.0

413840 rows × 6 columns

```
data.columns
data['Product Name'].value_counts()
```

Product Name	count
Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP Camera, White (Certified Refurbished)	1451
Apple MF259LL/A - iPhone 4s 8GB / 8MP Camera - Unlocked - Black (Certified Refurbished)	1241
BLU Studio 5.0 C HD Unlocked Cellphone, Black	1194
OtterBox Iphone 5/5S/SE Defender Case w/ Drop and Dust Proctection - Realtree AP Pink	1129
Motorola Moto E (1st Generation) - Black - 4 GB - Global GSM Unlocked Phone	1127
...	...
Nokia 8850 Unlock Cellular Phone (Silver)	1
Nokia 8850	1
Nokia 8310	1
Nokia 8210 Unlocked Cell Phone (Blue)	1
Nokia 8210 Unlocked Cell Phone (Black)	1

4410 rows × 1 columns

dtype: int64

```
[9]
Apple = data[data['Product Name'] == 'Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP Camera, White (Certified Refurbished)' ]
Apple
```

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes
16429	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	good buy	0.0
16430	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	great phonegood price	0.0
16431	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought and send to Africa the two are not wo...	0.0
16432	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	4	Had difficulty connecting to wf	0.0
16433	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	very good	0.0
...	...	...	...	...	...	...
17875	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought an unlocked phone but when I want to ...	0.0
17876	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	3	Video had no sound	0.0
17877	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	This products comes with problems on the speak...	NaN
17878	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	So it's been a month since I bought this produ...	0.0
17879	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	In pristine condition. I went to the Apple Sto...	NaN

1451 rows × 6 columns

```
[10]
print("Rows:",Apple.shape[0])
print("Columns:",Apple.shape[1])

Rows: 1451
Columns: 6
```

```
[11]
Apple["Review_Length"] = Apple["Reviews"].astype(str).apply(len)
Apple

/tmp/ipython-input-3866279548.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  Apple["Review_Length"] = Apple["Reviews"].astype(str).apply(len)
```

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes	Review_Length
16429	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	good buy	0.0	8
16430	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	great phonegood price	0.0	21
16431	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought and send to Africa the two are not wo...	0.0	51
16432	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	4	Had difficulty connecting to wf	0.0	31
16433	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	very good	0.0	9
...	...	...	...	...	...	...	...
17875	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought an unlocked phone but when I want to ...	0.0	83
17876	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	3	Video had no sound	0.0	18
17877	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	This products comes with problems on the speak...	NaN	124
17878	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	So it's been a month since I bought this produ...	0.0	348
17879	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	In pristine condition. I went to the Apple Sto...	NaN	217

1451 rows × 7 columns

```
[12]
shortest = Apple["Review_Length"].min()
longest = Apple["Review_Length"].max()
average = Apple["Review_Length"].mean()

print("Shortest Review Length:", shortest)
print("Longest Review Length:", longest)
print("Average Review Length:", average)

Shortest Review Length: 1
Longest Review Length: 1465
Average Review Length: 107.38042729152309
```

figure (3): Task3

Task 4:

```
[13]
#task4
Apple["Sentiment"] = Apple["Reviews"].astype(str).apply(sentimentAnalyzer)
Apple
```

```
/tmp/ipython-input-3870075984.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  Apple["Sentiment"] = Apple["Reviews"].astype(str).apply(sentimentAnalyzer)
```

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes	Review_Length	Sentiment
16429	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	good buy	0.0	8	Positive
16430	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	great phonegood price	0.0	21	Positive
16431	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought and send to Africa the two are not wo...	0.0	51	Neutral
16432	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	4	Had difficulty connecting to wf	0.0	31	Neutral
16433	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	very good	0.0	9	Positive
...	...	...	...	...	...	...	...	...
17875	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought an unlocked phone but when I want to ...	0.0	83	Neutral
17876	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	3	Video had no sound	0.0	18	Neutral
17877	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	This products comes with problems on the speak...	NaN	124	Negative

figure (4): Task4

Task 5:

```
[14]
#task5
sentiment_counts = Apple["Sentiment"].value_counts()
sentiment_counts
```

Sentiment	count
Positive	768
Neutral	573
Negative	110

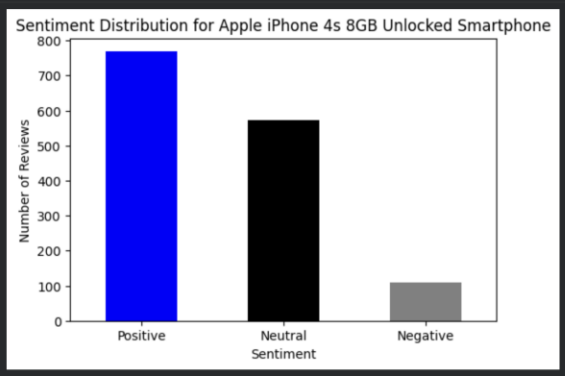
dtype: int64

```
dtype: int64
```

[15]

```python
import matplotlib.pyplot as plt

plt.figure(figsize=(6,4))
sentiment_counts.plot(kind="bar", color=["blue","black","gray"])
plt.title("Sentiment Distribution for Apple iPhone 4s 8GB Unlocked Smartphone")
plt.xlabel("Sentiment")
plt.ylabel("Number of Reviews")
plt.xticks(rotation=0)
plt.show()
```



[16]

```python
Apple["Polarity"] = Apple["Reviews"].astype(str).apply(lambda x: TextBlob(x).sentiment.polarity)
Apple
```

```
/tmp/ipython-input-415228445.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  Apple["Polarity"] = Apple["Reviews"].astype(str).apply(lambda x: TextBlob(x).sentiment.polarity)
```

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes	Review_Length	Sentiment	Polarity
16429	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	good buy	0.0	8	Positive	0.700000
16430	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	great phonegood price	0.0	21	Positive	0.800000
16431	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought and send to Africa the two are not wo...	0.0	51	Neutral	0.000000
16432	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	4	Had difficulty connecting to wf	0.0	31	Neutral	0.000000
16433	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	very good	0.0	9	Positive	0.910000
...	...	...	...	...	...	...	...	...	...
17875	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought an unlocked phone but when I want to ...	0.0	83	Neutral	0.000000
17876	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	3	Video had no sound	0.0	18	Neutral	-0.200000
17877	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	This products comes with problems on the speak...	NaN	124	Negative	-0.211667

[17]

```python
Examples_of_reviews = Apple.groupby("Sentiment").first()[["Reviews", "Polarity"]]
Examples_of_reviews
```

Sentiment	Reviews	Polarity
Negative	Bad batery	-0.7
Neutral	I bought and send to Africa the two are not wo...	0.0
Positive	good buy	0.7

```
[18]
neg_reviews = Apple[Apple["Sentiment"]=="Negative"]["Reviews"].astype(str)
neg_words = neg_reviews.str.lower().str.split()
neg_words_exploded = neg_words.explode()
neg_words_filtered = neg_words_exploded[neg_words_exploded.str.len() > 3]
top_neg_words = neg_words_filtered.value_counts().head(20)
top_neg_words
```

Reviews	count
this	56
phone	55
with	34
battery	24
very	24
after	17
it's	16
have	15
that	14
because	13
bought	12
return	11
will	11
from	11
product	10

```
[19]
from nltk.corpus import stopwords
import nltk

nltk.download('stopwords')
neg_reviews = Apple[Apple["Sentiment"]=="Negative"]["Reviews"].astype(str)
neg_words = neg_reviews.str.lower().str.split()
neg_words = neg_words.explode()
stop_words = set(stopwords.words("english"))
neg_words_clean = neg_words[(neg_words.str.len() > 3) & (~neg_words.isin(stop_words))]
neg_top_words = neg_words_clean.value_counts().head(20)
neg_top_words
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
```

Reviews	count
phone	55
battery	24
bought	12
return	11
product	10
phone.	10
work	9
slow	8
would	8
activate	8

```
# Many customers reported that the phone stopped working or failed shortly after purchase
# A large number of reviews complained about the battery draining quickly or not holding a charge.
# Many users mentioned returning the product, showing strong dissatisfaction.
# Customers frequently described the phone as slow, laggy, or unresponsive.
# Several reviews reported issues with the screen not working or malfunctioning.
# Some customers stated that the phone was not properly unlocked and did not work with their carriers.
```

```
wrong = Apple[(Apple["Rating"] <= 2) & (Apple["Sentiment"] != "Negative")]

wrong[["Reviews", "Polarity", "Sentiment", "Rating"]].head(30)
```

	Reviews	Polarity	Sentiment	Rating
16431	I bought and send to Africa the two are not wo...	0.000000	Neutral	1
16436	Do not buy this phone. it is used, no head pho...	-0.187500	Neutral	1
16439	sometimes the sound on videos , music & keyboa...	0.199074	Neutral	2
16440	Wifi couldnt be enabled. Apparently had a hard...	0.075000	Neutral	1
16463	Is old phone no is new I want my money is back...	0.078788	Neutral	1
16466	I was using an iPhone 6, but it was lost to th...	-0.136269	Neutral	1
16470	This product said it was for Verizon,but it wa...	0.000000	Neutral	1
16471	Phone is not charging, I bought it for my dad ...	-0.120000	Neutral	1
16472	Its was not unlocked I tried to put cricket wi...	0.000000	Neutral	1
16474	Not GSM!!	0.000000	Neutral	1
16483	It would in the area where I live.	0.136364	Neutral	2
16486	Doesn't have enough space. I only have about 3...	0.000000	Neutral	2
16488	The Battery is not lasting for more than two h...	0.250000	Positive	1
16501	Unlocked. See the attacked	0.000000	Neutral	1
16502	It is not unlocked, i love my virgin mobile ip...	0.222727	Positive	1
16504	Received the phone on the date promised and lo...	0.107879	Neutral	1
16507	Battery die	0.000000	Neutral	1

figure (5): Task5

Task 6:

```
#task6
Apple.to_csv("Apple.csv", index=False)
Apple
```

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes	Review_Length	Sentiment	Polarity
16429	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	good buy	0.0	8	Positive	0.700000
16430	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	great phonegood price	0.0	21	Positive	0.800000
16431	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought and send to Africa the two are not wo...	0.0	51	Neutral	0.000000
16432	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	4	Had difficulty connecting to wf	0.0	31	Neutral	0.000000
16433	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	very good	0.0	9	Positive	0.910000
...	...	...	...	...	...	...	...	...	...
17875	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought an unlocked phone but when I want to ...	0.0	83	Neutral	0.000000
17876	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	3	Video had no sound	0.0	18	Neutral	-0.200000
17877	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	This products comes with problems on the speak...	NaN	124	Negative	-0.211667
17878	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	So it's been a month since I bought this produ...	0.0	348	Neutral	0.141250
17879	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	In pristine condition. I went to the Apple Sto...	NaN	217	Positive	0.300000

figure (6): Task6

Use case

- **Product name :** Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP Camera, White (Certified Refurbished)
- **Number of rows** : 1451
- **Number of columns** : 6
- **Length of the shortest review** : 1
- **length of the longest review**: 1465
- **average length of the review**: 107.38042729152309

visualization and summative statistics

	Product Name	Brand Name	Price	Rating	Reviews	Review Votes	Review_Length	Sentiment	Polarity
16429	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	good buy	0.0	8	Positive	0.700000
16430	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	great phonegood price	0.0	21	Positive	0.800000
16431	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought and send to Africa the two are not wo...	0.0	51	Neutral	0.000000
16432	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	4	Had difficulty connecting to wf	0.0	31	Neutral	0.000000
16433	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	very good	0.0	9	Positive	0.910000
...	...	...	...	...	...	...	...	...	...
17875	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	I bought an unlocked phone but when I want to ...	0.0	83	Neutral	0.000000
17876	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	3	Video had no sound	0.0	18	Neutral	-0.200000
17877	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	1	This products comes with problems on the speak...	NaN	124	Negative	-0.211667
17878	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	So it's been a month since I bought this produ...	0.0	348	Neutral	0.141250
17879	Apple iPhone 4s 8GB Unlocked Smartphone w/ 8MP...	NaN	99.99	5	In pristine condition. I went to the Apple Sto...	NaN	217	Positive	0.300000

1451 rows × 9 columns

figure (7): sentiment labels and polarity scores

The figure shows a sample of reviews for the Apple iPhone 4s 8GB Unlocked Smartphone, along with their sentiment labels and polarity scores

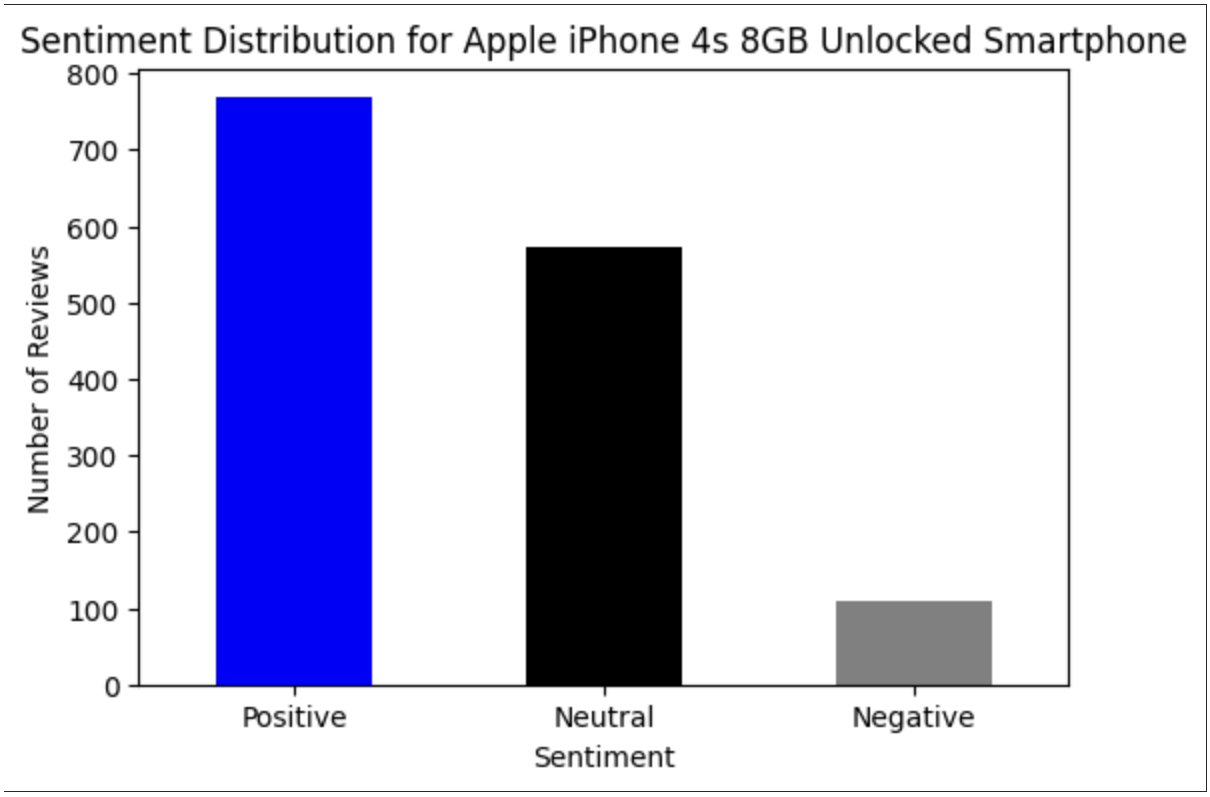


figure (8): bar chart

The bar chart clearly illustrates the sentiment distribution across 750+ reviews for this product:

- Positive Sentiment: The majority of the reviews are positive, with over **750** reviews expressing satisfaction with the product. These customers found the product reliable, with a high level of appreciation for its features such as battery life and ease of use.

- Neutral Sentiment: Around **580** reviews are neutral, where customers share mixed or indifferent opinions about their purchase, pointing out issues but not strongly negative.

- Negative Sentiment: There are only about **100** negative reviews, which are the smallest segment of the dataset. Customers with negative reviews generally cited issues such as bad battery performance or connectivity problems.

Sentiment	Reviews	Polarity
Negative	Bad batery	-0.7
Neutral	I bought and send to Africa the two are not wo...	0.0
Positive	good buy	0.7

figure (9): reviews

The table below includes several examples of the customers' reviews.

1. **Positive Sentiment Example:**

- Review: "good buy"
- Polarity: 0.7 (Positive)
- Analysis: The customer expresses satisfaction with the purchase, indicating that they consider it a good value.

2. **Neutral Sentiment Example:**

- Review: "I bought and sent to Africa the two are not working"
- Polarity: 0.0 (Neutral)
- Analysis: The review is neutral as it does not convey a strong emotion either way but highlights some functional issues with the product.

3. **Negative Sentiment Example:**

- Review: "Bad batery"
- Polarity: -0.7 (Negative)
- Analysis: The reviewer is dissatisfied with the product due to the poor battery life, resulting in a negative emotional rating..

Alignment between AI solution and client's needs

1. **Understanding Customer Opinions Accurately:**

- The AI solution categorizes customer reviews into sentiment categories—positive, neutral, or negative—using sentiment polarity scores. This enables the Amazon seller to gain accurate insights into customer emotions, highlighting what customers like and dislike about the product.

2. **Identifying the Most Common Problems in the Product:**

- The analysis of review text highlights the most frequently mentioned issues, such as battery problems (24 mentions), slow performance (8 mentions), and product malfunctions (such as the phone not working or not being properly unlocked). The word cloud or keyword counts, such as "phone" (55 mentions) and "battery" (24 mentions), help the seller easily identify the key issues customers face, such as hardware failures, poor battery life, and functionality issues.

3. **Improving Product Quality:**

- By pinpointing recurring problems like battery drain, poor functionality, and slow performance, the AI solution provides clear guidance for the seller to focus on areas for improvement. This can lead to more robust product designs, better quality control, and a better customer experience.

4. **Reducing Negative Feedback:**

- The sentiment analysis categorizes negative feedback and points out the specific complaints, enabling the seller to take corrective actions. Addressing common issues like malfunctioning batteries and unlocking problems can help reduce the number of negative reviews and improve overall customer satisfaction.

5. **Making Data-Driven Decisions**

- The AI solution provides data-driven insights by analyzing the frequency of keywords and sentiment polarity in reviews. For example, "battery" and "return" appear frequently in negative reviews, showing where the most dissatisfaction lies. These insights help the seller make informed decisions about product improvements, inventory management, and customer service strategies.

The solution developed in this project includes:

- A sentiment Analyzer function
- Review text analysis
- Insights from Negative Reviews

Reviews	
phone	55
battery	24
bought	12
return	11
product	10
phone.	10
work	9
slow	8
would	8
activate	8
iphone	7
disappointed	7
working.	7
even	7
good	7
screen	7
unlocked	7
life	7
know	7
compatible	7

figure (10): reviews

- Many customers reported that the phone stopped working or failed shortly after purchase
- A large number of reviews complained about the battery draining quickly or not holding a charge.
- Many users mentioned returning the product, showing strong dissatisfaction.
- Customers frequently described the phone as slow, laggy, or unresponsive.
- Several reviews reported issues with the screen not working or malfunctioning.
- Some customers stated that the phone was not properly unlocked and did not work with their carriers

Misclassified Reviews:

	Reviews	Polarity	Sentiment	Rating
16431	I bought and send to Africa the two are not wo...	0.000000	Neutral	1
16436	Do not buy this phone. it is used, no head pho...	-0.187500	Neutral	1
16439	sometimes the sound on videos , music & keyboa...	0.199074	Neutral	2
16440	Wifi couldnt be enabled. Apparently had a hard...	0.075000	Neutral	1
16463	Is old phone no is new I want my money is back...	0.078788	Neutral	1
16466	I was using an iPhone 6, but it was lost to th...	-0.136269	Neutral	1
16470	This product said it was for Verizon,but it wa...	0.000000	Neutral	1
16471	Phone is not charging, I bought it for my dad ...	-0.120000	Neutral	1
16472	Its was not unlocked I tried to put cricket wi...	0.000000	Neutral	1
16474	Not GSM!!	0.000000	Neutral	1
16483	It would in the area where I live.	0.136364	Neutral	2
16486	Doesn't have enough space. I only have about 3...	0.000000	Neutral	2
16488	The Battery is not lasting for more than two h...	0.250000	Positive	1
16501	Unlocked. See the attacked	0.000000	Neutral	1
16502	It is not unlocked, i love my virgin mobile ip...	0.222727	Positive	1
16504	Received the phone on the date promised and lo...	0.107879	Neutral	1
16507	Battery die	0.000000	Neutral	1
16508	Well this phone is breaking everyday like char...	-0.100000	Neutral	2
16514	The information for this item said it was BSM ...	0.425000	Positive	2
16526	Phone is in very good condition but the batter...	-0.001143	Neutral	1

figure (11): Misclassified reviews

Conclusion

To gain a deeper understanding of consumer perceptions of an Amazon product, sentiment analysis techniques were employed in this study. Reviews were automatically classified as positive, negative, or neutral using sentiment analysis functions and polarity scores, providing a clear view of user satisfaction levels. By analyzing the most frequently mentioned terms in negative reviews, key issues such as poor battery life, slow performance, and screen malfunctions were identified.

Overall, the AI solution proved effective in delivering valuable insights that can help the seller make informed, data-driven decisions to enhance product quality. The findings also highlight how sentiment analysis can support continuous product improvement by revealing hidden patterns in customer feedback. Future improvements could include the use of more advanced models to achieve even greater accuracy in sentiment analysis and offer even deeper insights into consumer behavior.